

Software Engineer

Essilfie Prince Mbeah

Relevant Links: [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Mobile](#) | [Credly Badges](#)

PROFILE

High-velocity Full-Stack & Backend Engineer specializing in robust system architectures (Spring Boot, NestJS, Rust) with an **AI-native approach to development**. Passionate about building highly efficient systems and developer tools, leveraging advanced AI coding agents to radically accelerate delivery cycles and engineering throughput.

EDUCATION

Kwame Nkrumah University of Science and Technology

November 2021 - November 2025

- BSc Computer Science with MTN Bright Scholarship Reloaded. First Class Honours

SKILLS

- Core Expertise:** Spring Boot, NestJS, Typescript, Rust, Java, PostgreSQL
- Working Knowledge:** Kotlin, Vue, NuxtJS, ReactJS, Angular, Flutter
- Tools:** Git, GitHub Actions, Docker, Cloudflare, GCP, WebSocket, Socket.io, Hurl, Claude Code, Opencode, AI Coding Agents

WORK EXPERIENCE

Amalitech

Nov 2025- Present

Backend Engineer - NSP

Takoradi, Ghana

- Worked in a team of 4 for 4 days to build a fully functional logging system for processing logs from different services and provided detailed analysis using Spring Boot and React
- Designed and implemented a JWT RSA-based authentication system with SSO integration and role-based access control using custom `@Authorization` annotations supporting five distinct user roles.
- Implemented a performance scoring engine used to manage PI periods, scoring frameworks, and automated target achievement calculations, reducing manual score computation effort.
- Integrated with an external GraphQL Resource Management System (RMS) via a scheduled sync service (Monday/Friday at 8am), resolving N+1 query issues with batch fetching.
- Supported developers with Angular implementations when the frontend fell behind schedule.
- Implemented aggregated monthly bonus computation per employee, seniority level management with pagination, and cascade deletion logic to maintain referential integrity across PI periods.
- Applied API-level rate limiting (100 req/min per user), enforced a one-active-framework database constraint via a PostgreSQL trigger, and maintained 85%+ line coverage with JaCoCo across unit and E2E (Hurl) test suites, passing SonarQube and Snyk security reports, utilizing AI pair-programming tools (Opencode/Claude) to accelerate test suite generation and edge-case discovery.

Zomujo Foundation

Jan 2025 – Feb 2026

Backend Engineer - Freelance ([Live Url](#))

Remote, Ghana

- Implemented cloud storage upload for images, labs, prescriptions and other files with google cloud API.
- Implemented Email and phone verification with Nodemailer and Cron Jobs.
- Designed and implemented prescription pdf generation with doctor's signature using pdf-lib
- Implemented consultation, labs, future visits reminders, email, including in-app notifications with Server Sent Events. Migrating EMR project from TSOA to NestJS with Prisma ORM. Implemented live communication with WebSocket and Socket.io

Zomujo Foundation

Oct 2024 – Dec 2024

Backend Engineer - Paid Intern

Remote, Ghana

- Worked on Drug inventory management system for hospitals and Pharmacies
- Personally worked on drugs and its categories, stock management and adjustments, and orders.

Zomujo Foundation

Oct 2023- Jan 2024

Backend Engineer - Intern

Remote, Ghana

- Actively engaged in a team of 7 developing backend for health services.
- I was given an additional role as a Black Box Tester

PROJECTS

Pseudocode Interpreter | [VScode link](#) | [Github Code](#) | [Rust Interpreter](#) (1.7k+ installs)

2024

- Engineered a custom runtime engine and interpreter in Rust to parse and execute pseudocode syntax, implementing lexers, parsers, and ASTs to handle features like multi-conditional flows, loops, functions, and multidimensional arrays.
- Developed and published a companion VS Code extension (**Pseudo Runner**) utilizing TypeScript to bridge the developer environment with the Rust native runtime binary.
- Designed this developer tool to streamline the engineering experience for early-stage students, enabling seamless, localized execution of logical syntax directly from the IDE interface

NASM-Compiler-Linux | [Repo](#) | [VScode Link](#) (14k+ installs)

2023

- Engineered an automated compilation and linking pipeline in TypeScript to transform raw NASM assembly code into native Linux ELF binaries directly within the IDE interface.
- Replaces the manual linking and running of assembly codes with a click of a button